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Zhejiang GreenPower I&E Co., Ltd
Professional Power Equipment Provider

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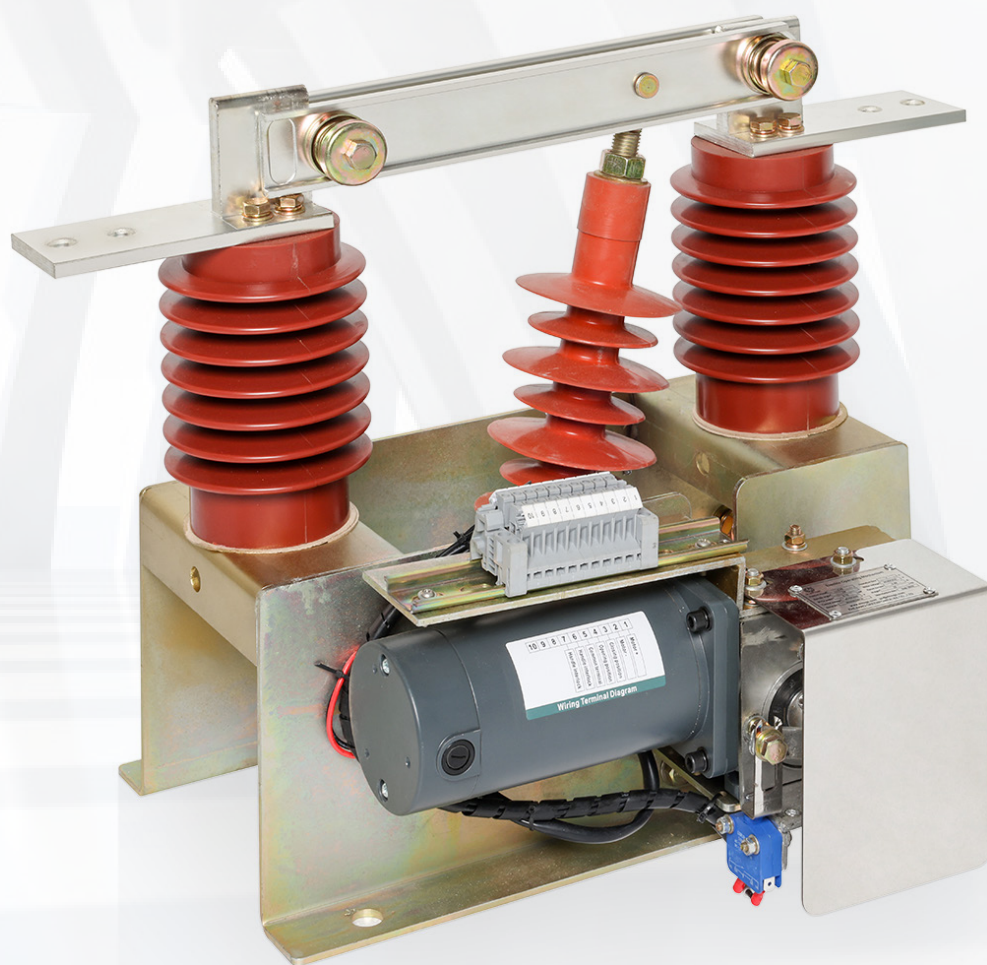
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GN19-12 Series

Indoor High Voltage Disconnect





About Us»

GreenPower promotes environmental awareness, and aims to create unprecedented happiness and wealth for our investors, employees, clients and partners. By focusing in the mid-to-high voltage and low voltage eld, and professionally working on R&D, manufacturing, marketing and service of high-end green intelligent switchgears, equipments and products.GreenPower sets to become a well-respected global company in the power industry.

GreenPower, is jointly established by a number of state-owned excellent industrial electrical professional manufacturers, we committed to meet the procurement needs of clients. It is a professional procurement service provider with entity of industrial electrical.

GreenPower provides ONE-STOP solutions service for all customers in the International area. It is located in the time-honored electrical industrial city, a collection of worldclass brand of industrial electrical products,excellent brand of domestic electrical products as the basis.

Innovative business philosophy, strong professional supply team, expert technical guidance, advanced Information network management platform, fast logistics, which makes GreenPower as your procurement expert by your side.

GreenPower adheres to its own brand and multi-brand integration, marketing dierentiated development strategy. The products involving low voltage, medium voltage and high voltage transmission and distribution products and industrial automation products.

It covering all aspects of distribution, logistics, warehousing, professional and technical engineering services, systems integration and complete sets of manufacturing. Stable business foundation for cooperation with customers and suppliers, and has established a good reputation to maintain its leading position in the market competition.

Human quests for transcendence, with transcendence we make progress, when the pursuit of unlimited become a belief, our dream began to realize.

Hard-working, dedicated GreenPower people, will be adhering to its past glory, beginners mind, let go sailing, to face the fierce competition in the future, and always help customers make the best choice, and strive to become the best and most reliable procurement service provider in the eld of industrial electrical.

01 Product overview

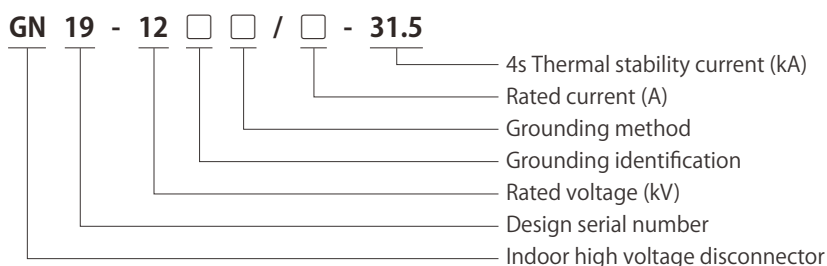
GN19-12 series indoor high voltage disconnectors are suitable for DC systems with a rated voltage of 12kV. They are used to disconnect and close circuits when voltage is present but no load current is present.

This series of products generally have a single pole electric operation for the main blade, and the knife switch adopts electric operation.

02 Environmental conditions and installation site for use

- Altitude not exceeding 1000m
- Indoor wind speed not exceeding 35m/s
- The ambient temperature shall not be higher than +40°C and not lower than -20°C
- The seismic intensity shall not exceed 8 degrees
- Not suitable for chemical substance locations with explosion and combustion hazards, severe vibrations and impacts, and severe corrosion of metals and insulation materials.

03 Product model and meaning



04 Execution standard

GB1985 "AC High Voltage Disconnectors and Earthing Switches"

GB11022 "General Technical Conditions for High Voltage Switchgear"

DL486 "Technical Conditions for Ordering AC High Voltage Disconnectors"

05 Main technical parameters

No.	Item		Unit	Data				
1	Rated voltage		kV	10				
2	Max working voltage		kV	12				
3	rated current		A	800	1000	1250	2000	2500
4	Rated short-time withstand current		kV	31.5	40	40	50	50
5	Rated peak withstand current		kV	80	100	100	125	125
6	Rated short-time withstand current time	Main knife switch	s	0.2				
		Grounding knife switch	s					
7	1min Power frequency withstand voltage (effective value)	to ground	kV	42				
		to break	kV	48				

06 Structural principle and installation and debugging

This is a single-pole disconnecter with an integrated electric mechanism and a common chassis. The switch consists of a main switch, a grounding switch, a moving contact, a rotating shaft, a tie rod insulator, a support insulator, and a motor mechanism. The single pole is connected to the motor mechanism through the rotating shaft. During operation, the rotating shaft drives the main switch to open and close through the tie rod insulator.

During debugging, place the switch on the foundation, let the main knife switch be in the middle position of opening and closing, connect the secondary control line, perform opening and closing operations, observe whether the opening and closing direction of the main knife is correct, if the opening and closing direction is incorrect, urgently remove the power supply or press the emergency stop button to prevent accidental opening and closing from damaging the disconnecter or electric mechanism. After the opening and closing are normal, observe whether the knife switch can stop normally after it is in place. If it fails to stop normally, urgently remove the power supply or press the emergency stop button, adjust the travel limit of the electric mechanism, adjust it to the appropriate position, and then perform the electric opening and closing operations. After the opening and closing are normal, perform the operation several times. After there is no problem, it can be operated normally.

07 Operation and inspection

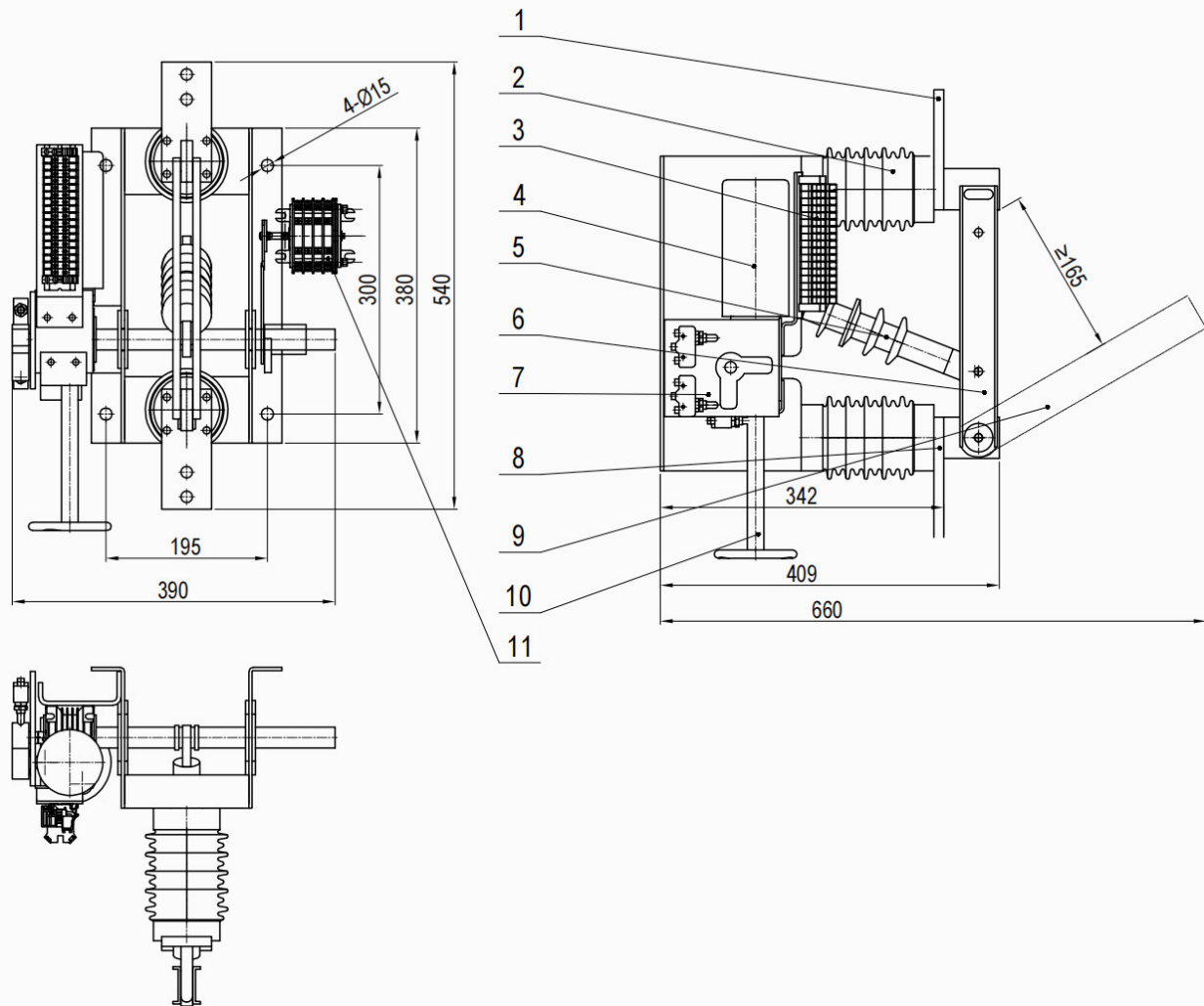
- **Inspection before putting into use:** The contact parts of the conductor should be in good contact, the connection with the busbar should be reliable, and the moving parts should be flexible.
- **Operation:** The switch must be operated after the line load is cut off.
- The switch should undergo a comprehensive maintenance once a year, which includes checking the conductive contact surface, lubricating the operating parts, replacing damaged parts, and surface rust prevention treatment.

08 Acceptance and storage

- **Acceptance:**
After the switch arrives at the destination, it should be opened and inspected as soon as possible. The product specifications should be checked according to the purchase contract, and the random attachments and documents should be checked according to the packing list. The product should be checked for serious damage. If any abnormal phenomena are found, please notify the manufacturer for joint negotiation and resolution.
- **Storage:**
After product acceptance, if not installed and used immediately, it should be stored in a covered and clean ventilated place, and checked every six months.

09 Ordering instructions

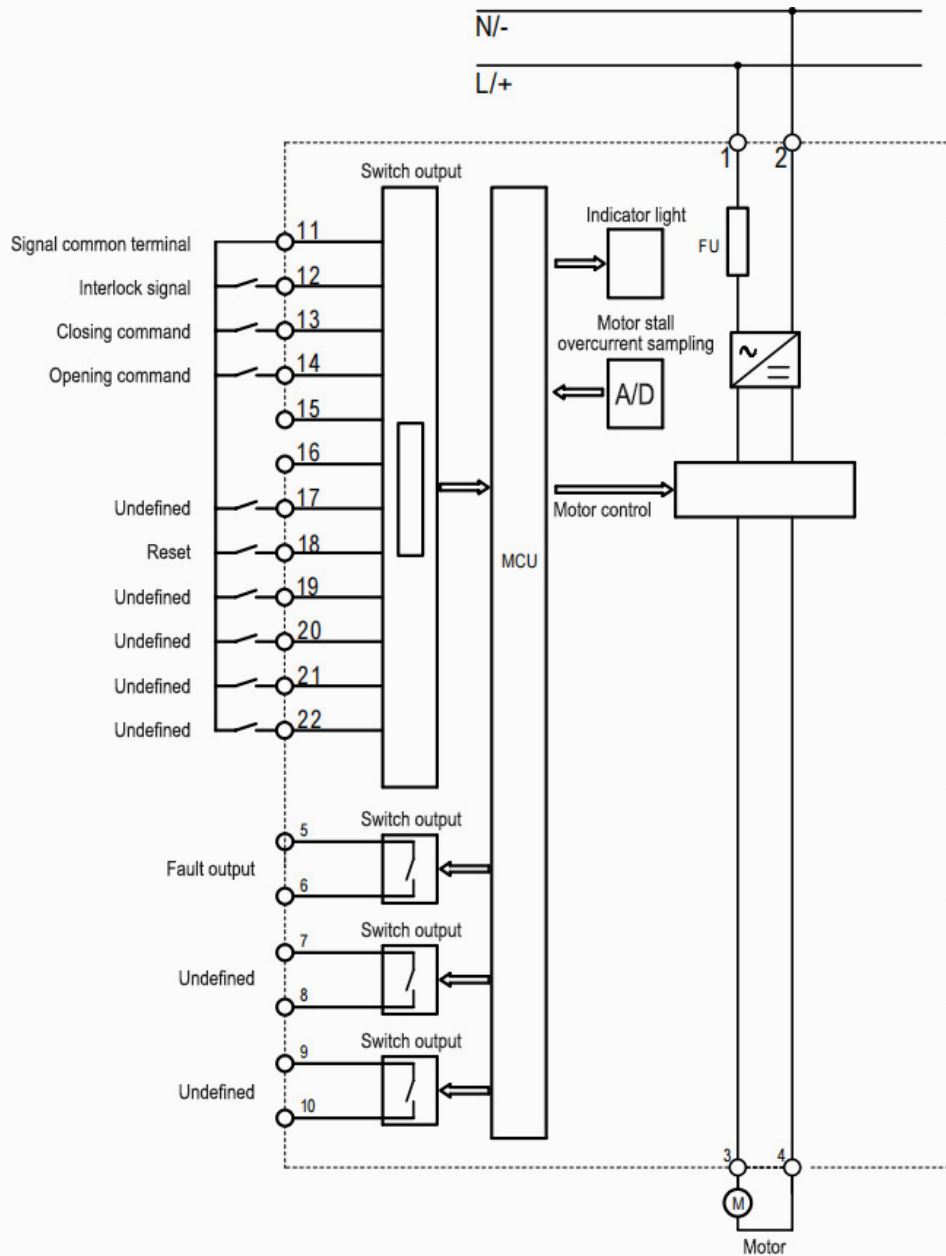
- Ordering should specify the model and specifications of the switch and the model of the operating mechanism.
- According to user needs, it can be processed into a single pole split or a two-phase common chassis type.
- There are two grounding forms: D1 dynamic contact grounding and D2 static contact grounding. Please specify them in the order.
- If there are special technical requirements, please indicate them together in the purchase contract.

10 GN□-12kV-800A single-pole electric disconnector


No.	Item	Qty.	Material
1	Fixed contact	1	T2Y
2	Insulator	2	Epoxy Resin
3	Terminal block	1	-
4	Motor	1	-
5	Pull rod	1	Epoxy Resin
6	Contact	2	-
7	Contact electric mechanism	1	-
8	Insulator	1	Epoxy Resin
9	Contact knife	2	T2Y
10	Operating handle	1	-
11	Auxiliary switch	1	-

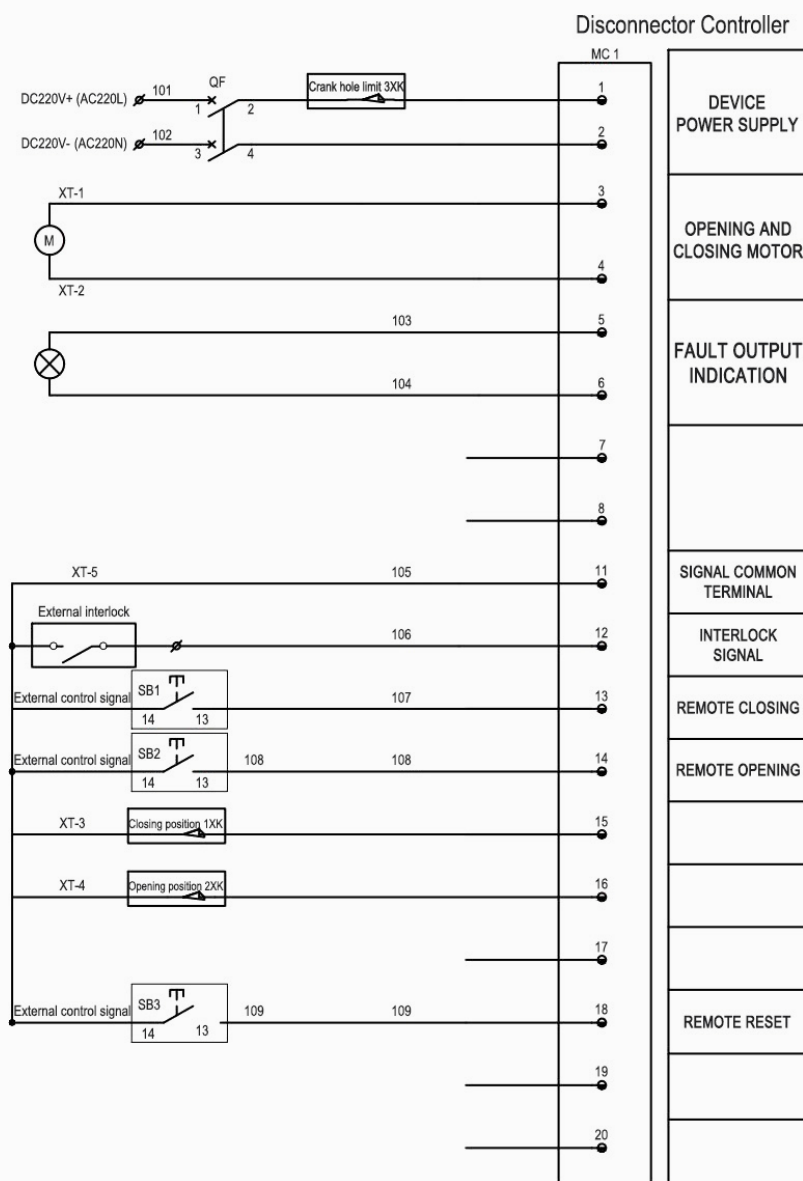
Item	Data
Rated voltage	12kV
Rated current	800A
Thermal stable current	31.5kA
Dynamic stable current	80kA
Power frequency withstand voltage	42kV/1min
Lightning impulse withstand voltage	75kV
Operation method	Electric and manual
Power supply voltage	AC/DC220V
Control voltage	AC/DC220V

11 Secondary schematic diagram of electric mechanism



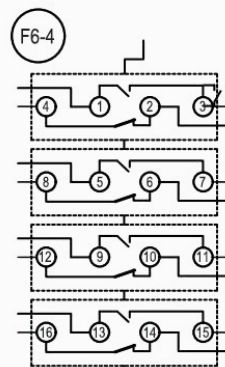
No.	Description	No.	Description	No.	Description	No.	Description
1	Power +	7	Undefined	13	Closing Command	19	Undefined
2	Power -	8	Undefined	14	Opening Command	20	Undefined
3	Motor +	9	Undefined	15	Closing Position	21	Undefined
4	Motor -	10	Undefined	16	Opening Position	22	Undefined
5	Fault Output	11	Signal Common Terminal	17	Undefined		
6	Fault Output	12	Interlock Signal	18	Reset		

12 Secondary schematic diagram of controller



Electric Mechanism Terminal

	XT
Motor +	1
Motor -	2
Closing position	3
Opening position	4
Signal common terminal	5
Handle interlock	6
Handle interlock	7
	8
	9
	10



F6-4 Auxiliary Switch Contact Diagram

13 Product overview of CJN series electric operating mechanism

13.1 Main application

The CJN series electric operating mechanism (hereinafter referred to as the mechanism) is mainly suitable for operating high-voltage disconnectors or earthing switches with rated voltages of 12kV and 40.5kV. The mechanism can perform opening and closing operations locally or remotely. The mechanism is also equipped with a manual operating handle for installation and commissioning, as well as operation when necessary.

13.2 Execution standard

GB1985-2004 "High Voltage AC Disconnectors and Earthing Switches"

GB/T11022-1999 "Common Technical Requirements for High Voltage Switchgear and Controlgear Standards"

DL/T486-2000 "Technical Conditions for Ordering AC High Voltage Disconnectors"

13.3 Normal operating environmental conditions

- Altitude not exceeding 1000m.
- The maximum air temperature shall not exceed 40°C , and the minimum air temperature shall not be lower than -40°C .
- The wind speed does not exceed 34m/s (equivalent to a wind pressure of 700Pa on the cylindrical surface).
- Ice level is class 10.
- The pollution level is class III

13.4 Main features

- The mechanism is designed to be installed inside the cabinet, and can also be installed integrally with the isolator.
- Control principle and on-site circuit breakers, grounding switches and other equipment can achieve electrical interlocking to ensure the correct coordination of operation timing, otherwise it will not be able to close or open the circuit.
- Operating voltage of the mechanism: AC220, DC220/110V, available for users to choose conveniently.

14 Main technical parameters

Mechanism model		CJN
Spindle angle		70°
Rated output torque		0.79 N.m
Opening and closing time		2~4 s
Control voltage		AC/DC 220V
Motor parameters	Model	GDM80/160-150-220-18S
	Power	0.15 kW
	Voltage	AC/DC 220V
	Speed	2100 r/min
	Rated current	0.68 A

15 Mechanical structure and control principle

15.1 Mechanical structure

- The main body of the mechanism is made of steel plates. The front of the institution is equipped with an operation panel.
- The mechanism adopts a structure that integrates a reducer and a motor. This structure has smooth transmission, low noise, low power consumption, particularly large output torque, and can ensure accurate opening and closing. The mechanical transmission of the mechanism is driven by an electric motor to reduce the speed of the gears. After the worm gear is reduced, the torque is transmitted to the output shaft, which drives the crank arm to perform opening and closing operations. The mechanism is equipped with dual protection of opening and closing electrical interlocking and mechanical limit, ensuring that the rotation angle of the spindle is limited to the accurate opening and closing position.
- The mechanism is equipped with a manual operating handle, which can be used for installation and debugging purposes, as well as necessary operations during power outages. When the handle is activated, it can automatically lock the electric operation to ensure the safety of the operator.

15.2 Control principle

To complete the opening and closing operation of the mechanism, it must have bidirectional operation function, which is achieved by dragging the motor to rotate in both directions. Please refer to the electrical schematic and wiring diagram for details.

● Operation control

The mechanism controller has closing, opening, and remote local operation buttons; The remote control operation buttons for closing, opening, and stopping should be provided by the user. Lift the button to activate the contactor and control the motor to rotate in both directions. The main shaft is driven by the mechanical reducer to rotate in both directions to complete the opening and closing of the circuit.

In the circuit diagram: 1XK and 2XK are limit switches for the opening and closing endpoints, respectively, which enable the mechanism to accurately stop at the endpoint after performing opening and closing operations. 3XK is the power switch inserted into the operating handle.

When the handle operation is enabled, inserting the handle into the motor shaft will automatically lock the electric operation, and manual operation can be directly performed. When restoring electric operation, the manual handle must be pulled out to unlock.

The controller panel has selection switches for local and remote operation.

● Interlock protection

In the figure, the interlocking signal is the auxiliary switch of the grounding switch and circuit breaker on site. Its normally closed contacts are connected in series in the control circuit. Only when the circuit breaker and grounding switch are in the open state, can the mechanism perform the opening and closing operation. Otherwise, the same applies. The timing interlock of this operation can limit misoperation after the switching operation. (If there is no interlock on site, it can be short circuited at the terminal block (factory short circuited, if there is an interlock switch, the short circuited wire can be removed)

● Status display

In the figure, FK is an auxiliary switch in the mechanism, which has an energy storage spring device. When the mechanism approaches the end position of opening and closing, the auxiliary switch quickly acts, and the indicator light changes accordingly, displaying opening or closing respectively. At the same time, the auxiliary switch also has many contacts available for users to choose from through wiring terminals.

● Overcurrent protection

In the picture, 1KD is a low-voltage circuit breaker used for motor and control circuits, providing overcurrent protection.

16 Transportation, acceptance and storage

16.1 Transportation

Wooden box packaging for institutions must not be rolled or inverted during transportation.

16.2 Acceptance

- Unpacking inspection: Check whether the box has any transportation damage and whether the nameplate is consistent with the order.
- Random attachments should be complete.
- Random technical documents should be complete.

16.3 Storage

It should be stored in a dry, well ventilated warehouse without harmful gases.

17 Installation, debugging and maintenance

17.1 Installation

- The transmission spindle of the mechanism and the operated disconnecter or earthing switch (hereinafter referred to as the switch) should be concentric.
- The transmission shaft between the main shaft and the switch is consistent with other components that are equipped.

17.2 Debugging

• No-load test

Insulation trial production: The low-voltage electrical equipment provided by the institution should meet the power frequency withstand voltage of 2000V.

Manual operation test: Use the handle to test the operation, and the mechanism should be flexible without any jamming phenomenon, and the auxiliary switch should switch normally.

Electric test: Connect the power supply, buttons, small circuit breakers, transfer switches, limit switches, auxiliary switches, and drag motors as specified in the diagram. The motor should operate normally and rotate freely in both directions without any abnormal noise. Accurate opening and closing of the switch, reliable locking.

• Load test

Make mechanical connections between the mechanism and the operated switch and tighten them securely.

Manual opening and closing test, using a handle to drive the mechanism, to perform closing and opening operations on the switch. Repeated operations should have no jamming phenomenon and be synchronized in place.

Electric operation test: Before the test, manually operate the switch to the middle position of opening and closing, connect the power supply, and observe the indicator light on the control panel. If the power indicator light is always on, it indicates that the operation is possible. If the indicator light is not on, adjust the phase sequence of the incoming power supply. After the light is on, turn on the closing or opening button, and then turn on the stop button to simulate a jog operation. Pay attention to the rotation of the spindle and confirm the correct direction of operation.

• Adjustment points

When manually closing and opening the switch, check the gap between the limit piece and the baffle when the end limit switch is just switched. From this position to the end position, the handle should be able to shake 4.8 ± 1.8 turns, and the DC mechanism handle should be able to shake 8 ± 1 turn.

17.3 Maintenance

- Maintenance personnel should understand the transmission mode, debugging methods, electrical control principles, and other technical aspects of the mechanism. The maintenance cycle should be conducted once a year, and any problems that arise during operation should be recorded. If necessary, the manufacturer can be notified.

• Precautions

Power outage maintenance. Do not replace components, including electrical components, that are different from the design of the mechanism.

If the power supply line is disassembled or replaced during maintenance, please repeat the motor rotation direction test according to the precautions.

If the handle has been used for operation, be sure to close the baffle, otherwise it cannot be electrically operated.

18 Random attachments and technical documents

18.1 Attachment

- Electric mechanism: 1 unit
- Handle: 1 unit

18.2 Technical documentation

- Product test report
- Installation and operation manual

19 Ordering instructions

- Explain the model, name, and quantity of the organization.
- Institutional control power supply voltage, AC or DC voltage level.
- Auxiliary switch pole number, if not specified, supply according to 8 poles.

[illegible]